

Credit Name: CSE 2120

Assignment Name: Exercise 15 Mankala

Error Log Entry

What error message did you encounter (if any)?

None

What unexpected behavior did your program exhibit?

program got stuck inside loop

What caused the issue? (e.g., syntax error, logic error, incorrect function usage, etc.)

logic error

Include a screenshot of specific lines of code.

```
100 |         if (heldStones == 1 && side == turn) {
101 |             switch(turn) {
102 |                 case PLAYER:
103 |                     if (board.getPlayerPit(currentPit) == 0) {
104 |
105 |                         board.updatePlayerBin(board.getOpponentPit(currentPit) + 1);
106 |                         board.setOpponentPit(currentPit, 0);
107 |
108 |                     }
109 |                     return;
110 |                 case OPPONENT:
111 |                     if (board.getOpponentPit(currentPit) == 0) {
112 |
113 |                         board.updateOpponentBin(board.getPlayerPit(currentPit) + 1);
114 |                         board.setPlayerPit(currentPit, 0);
115 |
116 |                     }
117 |             }
118 |         }
```

How did you fix the issue?

Moved return statement and added in break;

Provide the corrected code or solution using a screenshot.

```
105@         if (heldStones == 1 && side == turn) {
106@             switch(turn) {
107@                 case PLAYER:
108@                     if (board.getPlayerPit(currentPit) == 0) {
109@
110@                         board.updatePlayerBin(board.getOpponentPit(currentPit) + 1);
111@                         board.setOpponentPit(currentPit, 0);
112@
113@                     }
114@                     break;
115@                 case OPPONENT:
116@                     if (board.getOpponentPit(currentPit) == 0) {
117@
118@                         board.updateOpponentBin(board.getPlayerPit(currentPit) + 1);
119@                         board.setPlayerPit(currentPit, 0);
120@
121@                     }
122@                     break;
123@             }
124@         }
125@         return;
```

Error Log Entry

What error message did you encounter (if any)?

None

What unexpected behavior did your program exhibit?

Program kept skipping initial stone being added to pit

What caused the issue? (e.g., syntax error, logic error, incorrect function usage, etc.)

logic error

Include a screenshot of specific lines of code.

```
142@ void move(User turn, int startPit) {
143@
144@     int heldStones = 0;
145@     int currentPit = 0;
146@     User side = null;
147@     boolean playAgain = false;
148@     System.out.println("Worked line 146");
149@     switch(turn) { // Setting up what side and pit to start with
150@
151@     case PLAYER:
152@         heldStones = board.getPlayerPit(startPit);
153@         board.setPlayerPit(startPit, 0);
154@         currentPit = startPit + 1;
155@         side = PLAYER;
156@         break;
157@
158@     case OPPONENT:
159@         heldStones = board.getOpponentPit(startPit);
160@         board.setOpponentPit(startPit, 0);
161@         currentPit = startPit + 1;
162@         side = OPPONENT;
163@         break;
164@     }
165@     while(heldStones > 0) {
166@
167@         // Switching the sides ///
168@         if (currentPit == 6) { // Outside of the pits
169@             if (side == turn) { // Users turn, on Users side or vice versa
170@
171@                 switch(turn) {
172@                     case PLAYER:
173@                         board.updatePlayerBin(1);
174@                         System.out.println("Worked line 180");
175@                         if (heldStones == 1) { playAgain = true; }
176@                         heldStones -- 1;
177@                         break;
178@
179@                     case OPPONENT:
180@                         board.updateOpponentBin(1);
181@                         System.out.println("Worked line 187");
182@                         if (heldStones == 1) { playAgain = true; }
183@                         heldStones -- 1;
184@                         break;
185@                 }
186@
187@             }
188@             currentPit = 0;
189@             side = (side == PLAYER) ? OPPONENT : PLAYER;
190@         }
191@     }
192@ }
```

```

        if (heldStones == 1 && side == turn) {
            switch(turn) {
                case PLAYER:
                    if(board.getPlayerPit(currentPit) == 0) {
                        board.updatePlayerBin(board.getOpponentPit(currentPit) + 1);
                        board.setOpponentPit(currentPit, 0);
                    }
                    break;
                case OPPONENT:
                    if(board.getOpponentPit(currentPit) == 0) {
                        board.updateOpponentBin(board.getPlayerPit(currentPit) + 1);
                        board.setPlayerPit(currentPit, 0);
                    }
                    break;
            }
            return;
        }
    }
}

```

How did you fix the issue?

removed extra currentPit += 1;

Provide the corrected code or solution using a screenshot.

```

142 void move(User turn, int startPit) {
143
144     int heldStones = 0;
145     int currentPit = 0;
146     User side = null;
147     boolean playAgain = false;
148     // System.out.println("Worked line 146");
149     switch(turn) { // Setting up what side and pit to start with
150
151         case PLAYER:
152             heldStones = board.getPlayerPit(startPit);
153             board.setPlayerPit(startPit, 0);
154             currentPit = startPit + 1;
155             side = PLAYER;
156             break;
157
158         case OPPONENT:
159             heldStones = board.getOpponentPit(startPit);
160             board.setOpponentPit(startPit, 0);
161             currentPit = startPit + 1;
162             side = OPPONENT;
163             break;
164     }
165 }

```

Error Log Entry

What error message did you encounter (if any)?

none

What unexpected behavior did your program exhibit?

when taking from opponents pit if the added stone pit was empty if used the wrong pit

What caused the issue? (e.g., syntax error, logic error, incorrect function usage, etc.)

logci error

Include a screenshot of specific lines of code.

```

119     if (heldStones == 1 && side == turn) {
120         switch(turn) {
121             case PLAYER:
122                 if(board.getPlayerPit(currentPit) == 0) {
123                     board.updatePlayerBin(board.getOpponentPit(currentPit + 1) + 1);
124                     board.setOpponentPit(currentPit, 0);
125                     System.out.println("Worked line 211");
126                 }
127                 break;
128             case OPPONENT:
129                 if(board.getOpponentPit(currentPit) == 0) {
130                     board.updateOpponentBin(board.getPlayerPit(currentPit) + 1);
131                     board.setPlayerPit(currentPit, 0);
132                     System.out.println("Worked line 219");
133                 }
134                 break;
135         }
136     }
137 }

```

How did you fix the issue?

added in method to convert the values for the opponents side

Provide the corrected code or solution using a screenshot.

```

56 // Opponents Pit Difference
57 int OPD(int value) {
58     switch(value) {
59         case 2: return 2;
60         case 3: return 3;
61         default: return 5 - value;
62     }
63 }
64
65 void updateOpponentBin(int amount) { opponentBin += amount; }
66 void setOpponentPit(int pit, int amount) { opponentSide[OPD(pit)] = amount; }
67 void updateOpponentPit(int pit, int amount) { opponentSide[OPD(pit)] += amount; }
68 int getOpponentPit(int pit) { return opponentSide[OPD(pit)]; }
69

```

Error Log Entry

What error message did you encounter (if any)?

index out of bounds for length

What unexpected behavior did your program exhibit?

randomly quit during opponents turn

What caused the issue? (e.g., syntax error, logic error, incorrect function usage, etc.)

Include a screenshot of specific lines of code.

```

285 void opponentTurn() {
286
287     System.out.print("OPPONENT TURN");
288     board.getDisplay();
289     Random rand = new Random();
290     int move = 0;
291
292     while(move == 0) {
293         move = board.getOpponentPit(rand.nextInt(5));
294     }
295
296     move(OPPONENT, move);
297 }

```

How did you fix the issue?

Changed what the if loop check for and some logic

Provide the corrected code or solution using a screenshot.

```

285 void opponentTurn() {
286
287     System.out.print("OPPONENT TURN");
288     board.getDisplay();
289     Random rand = new Random();
290     int move = -1;
291
292     while(move < 0 || move > 5 || board.getOpponentPit(move) == 0) {
293         move = rand.nextInt(5);
294     }
295     move(OPPONENT, move);
296 }
297

```

Error Log Entry

What error message did you encounter (if any)?

none

What unexpected behavior did your program exhibit?

displaying the wrong title for each turn

What caused the issue? (e.g., syntax error, logic error, incorrect function usage, etc.)

Logic error

Include a screenshot of specific lines of code.

```

202
203 case PLAYER:
204     heldStones = board.getPlayerPit(startPit);
205     board.setPlayerPit(startPit, 0);
206     currentPit = startPit + 1;
207     nextTurn = OPPONENT;
208     break;
209
210 case OPPONENT:
211     heldStones = board.getOpponentPit(startPit);
212     board.setOpponentPit(startPit, 0);
213     currentPit = startPit + 1;
214     nextTurn = PLAYER;
215     break;
216 }
217

```

How did you fix the issue?

Switched the initial next turn decider

Provide the corrected code or solution using a screenshot.

```

203 case PLAYER:
204     heldStones = board.getPlayerPit(startPit);
205     board.setPlayerPit(startPit, 0);
206     currentPit = startPit + 1;
207     nextTurn = PLAYER;
208     break;
209
210 case OPPONENT:
211     heldStones = board.getOpponentPit(startPit);
212     board.setOpponentPit(startPit, 0);
213     currentPit = startPit + 1;
214     nextTurn = OPPONENT;
215     break;
216 }
217

```